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1. If the vertices of a triangles are $O(0, 0)$, $A(a, 0)$ and $B(0, a)$. Then, the distance between its circumcenter and orthocenter is:

- (a) $\frac{a}{2}$ (b) $\frac{a}{\sqrt{2}}$
(c) $\sqrt{2}a$ (d) $\frac{a}{4}$

MATHS – Coordinate Geometry
(CUET-PG 2022)

2. The straight line $x + y = 0$, $3x + y - 4 = 0$ and $x + 3y - 4 = 0$ form a triangle which is

- (a) Right angled
(b) Equilateral
(c) Isosceles
(d) Isosceles and right angled

MATHS – Straight Line (CUET-PG 2022)

3. If one of the lines of $ax^2 + 2hxy + by^2 = 0$ bisects the angle between the axes in the first quadrant, then

- (a) $h^2 - ab = 0$ (b) $h^2 + ab = 0$
(c) $(a + b)^2 = h^2$ (d) $(a + b)^2 = 4h^2$

MATHS – Pair of Lines (CUET-PG 2022)

4. Let E be the ellipse $\frac{x^2}{9} + \frac{y^2}{4} = 1$ and C be the circle $x^2 + y^2 = 9$. Let P and Q be the points (1, 2) and (2, 1) respectively. Then

- (a) Q lies inside C but outside E
(b) Q lies outside both C and E
(c) P lies inside both C and E
(d) P lies inside C but outside E

MATHS – Conic Section (Ellipse/Circle)
(CUET-PG 2023)

5. A triangle with vertices (4, 0), (-1, -1), (3, 5) is

- (a) Isosceles and right angled
(b) Isosceles but not right angled
(c) Right angled but not isosceles
(d) Neither right angled nor isosceles

MATHS – Coordinate Geometry (Triangle)
(CUET-PG 2023)

6. Match List I with List II

LIST I		LIST II	
(A)	The angle between the straight lines, $2x^2 + 3y^2 - 7xy = 0$ is	(i)	$\tan^{-1} \frac{3}{5}$

(B)	The circles $x^2 + y^2 + x + y = 0$ and $x^2 + y^2 + x - y = 0$ intersect at angle	(ii)	25π
(C)	The area of circle centred at (1, 2) and passing through (4, 6) is	(iii)	$\pi/4$
(D)	The parabolas $y^2 = 4x$ and $x^2 = 32y$ intersect at point (16, 8) at angle	(iv)	$\pi/2$

Choose the correct answer from the options given below:

- (a) A-IV, B-III, C-I, D-II (b) A-IV, B-III, C-II, D-I
(c) A-III, B-IV, C-II, D-I (d) A-III, B-IV, C-I, D-II

MATHS – Coordinate Geometry (CUET-PG 2023)

7. If a Chord which is normal to the parabola $y^2 = 4x$ at one end subtends a right angle at the vertex, then its slope is -

- (a) 1 (b) $\sqrt{3}$ (c) $\sqrt{2}$ (d) 4.2

MATHS – Parabola (Normal) (CUET-PG 2023)

8. A straight line has equation $y = -x + 6$ which of the following line is parallel to it?

- (a) $2y + 3x = -5$ (b) $-3x - 3y + 7 = 0$
(c) $2y = -x + 12$ (d) $y - x = \frac{1}{10}$

MATHS – Straight Line (CUET-PG 2023)

9. Given below are two statements: One is labelled as Assertion A and the other is labelled as Reason R.

Assertion A: If two circles intersect at two points, then the line joining their centres is perpendicular to the common chord

Reason R: The perpendicular bisectors of two chords of a circle intersect at its centre.

In the light of the above statements, choose the correct answer from the options given below:

- (a) Both A and R are true and R is the correct explanation of A
(b) Both A and R are true but R is not the correct explanation of A
(c) A is true but R is false
(d) A is false but R is true

MATHS – Circle Geometry (CUET-PG 2023)



10. The tangent to the hyperbola $x^2 - y^2 = 3$ are parallel to the straight line $2x + y + 8 = 0$ at the following points:

- (a) (2, 2), (1, 2) (b) (2, -1), (-2, 1)
(c) (-2, -1), (1, 2) (d) (-2, -1), (-1, -2)

MATHS – Hyperbola (CUET-PG 2023)

11. A circle S passes through the point (0, 1) and is orthogonal to the circles $(x - 1)^2 + y^2 = 16$ and $x^2 + y^2 = 1$. Then

- (A) Radius of S is 8 (B) Radius of S is 7
(C) Centre of S is (-7, 1) (D) Centre of S is (-8, 1)
(a) A only (b) A and B only
(c) B and C only (d) D only

MATHS – Circle (CUET-PG 2023)

12. Which of the following statement are TRUE?

- (A) A equation $ax^2 + bx + c = 0$ has real and distinct roots if $b^2 - 4ac \geq 0$ and $a \neq 0$
(B) The unit digit in 49^{18} is 1
(C) If two lines make complementary angles with the axis of x then the product of their slopes is 1.
(D) The line $bx - ay = 0$ meet the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$

Choose the correct answer from the options given below

- (a) A and D only (b) B and C only
(c) A, B and C only (d) A, B and D only

Algebra + Coordinate Geometry (Mixed Concept MCQ) (CUET-PG 2024)

13. Match List-I with List-II

List-I		List-II	
A.	Eccentricity of the conic $X^2 - 4x + 4y + 4y^2 = 12$ is	I.	10/3
B.	Latus rectum of conic $5x^2 + 9y^2 = 45$ is	II.	1
C	The straight line $x + y = a$ touches the curve $y = x - x^2$ then value of a is	III.	2
D.	Eccentricity of conic $3x^2 - y^2 = 4$ is	IV.	$\sqrt{3}/2$

Choose the correct answer from the options given below

- (a) A-I, B-II, C-IV, D, III (b) A-II, B-I, C-III, D-IV
(c) A-IV, B-I, C-II, D-III (d) A-IV, B-II, C-I, D-III

Conic Sections – Ellipse & Hyperbola Properties (CUET-PG 2024)

14. A line passes through a point (2,3) such that sum of its intercepts on the axes is 12 then equation of line (/s) is (/are) given by

- (A) $3x + y = 9$ (B) $x + 3y = 9$
(C) $x + 2y = 8$ (D) $5x + 7y = 35$

Choose the correct answer from the options given below

- (a) A only (b) A, B and C only
(c) A and C only (d) B, C and D only

Straight Lines – Intercept Form (CUET-PG 2024)

15. The equation of a circle that passes through the points (3,0) and (0,-2) and its centre lies on a line $2x + 3y = 3$ then equation of the circle is given by

- (a) $x^2 + y^2 + 2x + 16y + 72 = 0$
(b) $10x^2 + 10y^2 - 6x - 16y - 72 = 0$
(c) $5x^2 + 5y^2 + 6x + 16y + 72 = 0$
(d) $10x^2 + 10y^2 + 6x + 16y - 72 = 0$

Circle – Equation of Circle (CUET-PG 2024)

16. If the vertices of a triangle are (1,2), (2,5) and (4,3) then the area of the triangle is

- (a) 3 sq. units (b) 4 sq. units
(c) 6 sq. units (d) 8 sq. units

Coordinate Geometry – Area of Triangle (CUET-PG 2024)

17. An equilateral triangle is inscribed in a parabola $y^2 = 8x$ whose one vertex is at the vertex of the parabola then the length of the side of the triangle is:

- (a) $8\sqrt{3}$ units (b) $16\sqrt{3}$ units
(c) $4\sqrt{3}$ units (d) $\sqrt{3}/2$ units

Parabola – Geometry Application (CUET-PG 2024)

18. A equation of a conic is $ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$, where a, b, c, f, g and h are constants Then which of the following statement are true

- (A) The given conic is circle if $a = 0$ and $b = 0$
(B) The given conic is circle if $a = b \neq 0$ and $h = 0$
(C) The given conic is circle if $a = b \neq 0$ and $h \neq 0$
(D) The given conic represents a pair of real and distincts straight lines if $f = g = c = 0$ and $h^2 - ab > 0$

Choose the correct answer from the options given below

Conic Sections – General Second Degree Equation (CUET-PG 2024)

- (a) B only (b) B and D only
(c) A, B C and D (d) D only



19. The points $(K, 2 - 2K)$, $(-K + 1, 2K)$ and $(-4 - k, 6 - 2K)$ are collinear if:

(CUET-PG 2025) Co-ordinate

- (A) $K = \frac{1}{2}$ (B) $K = \frac{-1}{2}$
(C) $K = \frac{3}{2}$ (D) $K = -1$
(E) $K = 1$

Choose the **correct** answer from the options given below:

- (a) (A) and (D) only (b) (A) and (E) only
(c) (B) and (D) only (d) (D) only

20. The length of major axis and coordinate of vertices for the ellipse $3x^2 + 2y^2 = 6$ respectively are:

(CUET-PG 2025) Ellipse

- (a) $2\sqrt{2}$, $(0, \pm\sqrt{3})$
(b) $2\sqrt{3}$, $(0, \pm\sqrt{3})$
(c) $2\sqrt{2}$, $(\pm\sqrt{3}, 0)$
(d) $2\sqrt{3}$, $(\pm\sqrt{3}, 0)$

21. If $x^2 = -16y$ is an equation of parabola then:

- (A) directrix is $y = 4$
(B) directrix is $x = 4$
(C) co-ordinates of focus are $(0, -4)$
(D) co-ordinates of focus are $(-4, -0)$

- (E) length of latusrectum = 16

(CUET-PG 2025) Parabola

Choose the correct answer from the options given below:

- (a) (A) and (E) only
(b) (B), (C) and (E) only
(c) (A), (C) and (E) only
(d) (B), (D) and (E) only

22. The center and radius for the circle $x^2 + y^2 + 6x - 4y + 4 = 0$ respectively are:

(CUET-PG 2025) Circle

- (a) (2, 3) and 3 (b) (3, 2) and 8
(c) (2, -3) and 3 (d) (-3, 2) and 3

ANSWER KEY

1.	2.	3.	4.	5.	6.	7.	8.	9.	10.
b	c	d	d	a	c	c	b	a	b
11.	12.	13.	14.	15.	16.	17.	18.	19.	20.
c	b	c	c	b	b	b	b	a	b
21.	22.								
c	d								